

# **bToTEM model of the Bank of Canada for projection and policy analysis**

Serguei Maliar

June 6

PKU-Zurich PhD Summer School on  
Machine Learning for Macroeconomics and Finance  
July 6–10, 2026 · Beijing, China

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# Paper

## **"When the U.S. catches a cold, Canada sneezes: a lower-bound tale told by deep learning"**

by Vadym Lepetyuk, Lilia Maliar and Serguei Maliar

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# Large-scale DSGE models for projection and policy analysis

Mostly used by central banks and government agencies:

- International Monetary Fund's Global Economy Model, GEM (Bayoumi et al., 2001);
- US Federal Reserve Board's SIGMA model (Erceg et al., 2006);
- Bank of Canada Terms of Trade Economic Model, ToTEM II (Dorich et al. 2013);
- European Central Bank's New Area-Wide Model, NAWM (Coenen et al. 2008);
- Bank of England COMPASS model (Burgess et al., 2013);
- Swedish Riksbank's Ramses II model (Adolfson et al., 2013).

# What are central bank models aimed to do?

1. Central bank models must mimic as close as possible the actual economies in every possible dimension.
  - *Then, the policymakers can produce realistic simulation of alternative policies and to choose the best one*
2. Central bank models must be rich and flexible enough to describe interactions between many variables of interest, including different types of foreign and domestic consumption, investment, capital, labor, prices, exchange rate, etc.
  - *Central bank models may contain hundreds of equations*
  - *Their estimation, calibration, solution and simulation are highly nontrivial tasks*
3. Central banks need DSGE models for policy analysis.
  - *Econometric models have limitations for policy analysis (Lucas critique)*

# Linearized solutions to central bank models

- The central banks use linear (first-order) perturbation methods
  - dynare, IRIS.
- *Advantages:*
  - computationally inexpensive;
  - simple to use;
  - can be applied to very large problems.
- *Potential drawbacks:*
  - insufficiently accurate in the presence of strong nonlinearities;
  - does not allow to study occasionally binding constraints (with an exception of the linear case; see OccBin of Guerrieri and Iacoviello, 2015, and IRIS).
- *Nonlinear effects can be economically significant:*
  - Approximation errors in the new Keynesian model can reach hundreds percent near ZLB; see Judd, Maliar and Maliar (Econometrica, 2017).

# The view of nonlinearity before Great Recession

Policymakers were not concerned about nonlinearities before Great Recession.

- Leahy (2013): *“Prior to the crisis, it was easier to defend the proposition that non-linearities were unimportant than it was to defend the proposition that non-linearities were essential for understanding macroeconomic dynamics.”*
- Bullard (2013): *“... the idea that U.S. policymakers should worry about the nonlinearity of the Taylor-type rule and its implications is sometimes viewed as an amusing bit of theory without real ramifications. Linear models tell you everything you need to know. And so, from the denial point of view, we can stick with our linear models...”*

# The view of nonlinearity after Great Recession

Policymakers were confronted with questions after Great Recession.

- *Macroeconomists overlooked Great Recession. Why?*
  - Is it because they had bad models?
  - Or is it because they used bad linear methods that distorted policy implications of good models?
- In the light of these questions, Bank of Canada created a working group whose objective was to construct a global solution to its large-scale DSDE model which led to the present paper.



# Methodological contribution

- Apply deep supervised and unsupervised learning techniques to break the curse of dimensionality and to construct accurate global nonlinear solutions to large-scale central bank models.
- Quantify the difference between local linear and global nonlinear solutions in realistically calibrated central bank models.
- Explore whether inaccuracies of the linearization method can significantly distort policy implications of central bank models.
- Establish the key determinants of the recent ZLB episode in the Canadian economy.
- Analyze policies that would allow to avoid the ZLB crisis in the future.

# Empirical findings

1. We demonstrate that the international transmission of ELB is empirically plausible mechanism for explaining the Canadian ELB experience.
2. We show that it is relatively easy to generate realistic ELB (or ZLB) episodes in new Keynesian models via the foreign shocks unlike the domestic shocks.
3. We find that the Canadian economy would entirely avoid the ELB episode if the target inflation rate were 3 instead of 2 percent.
4. Contrary to what we expected, we find that the ELB constraint plays a relatively minor role in the bToTEM's performance.
5. We discover that other nonlinearities – those not associated with ELB – can play an important role in the model's predictions such as the uncertainty effect.
6. Most strikingly, we find that the closing condition, used to ensure stationarity in open-economy models, plays an important role in the bToTEM dynamics. This is at odds with well known result of Schmitt-Grohé and Uribe (2003) because we consider nonlinear model while they focus on linearization.

# Outline of the talk

1. Great Recession in Canada.
2. Bank of Canada ToTEM model for projections and policy analysis.
3. Breaking the curse of dimensionality using machine learning:
  - *clustering analysis*;
  - *neural networks and deep learning*.
4. Policy experiments:  
*understanding the Canadian ZLB crisis.*
5. Conclusion.

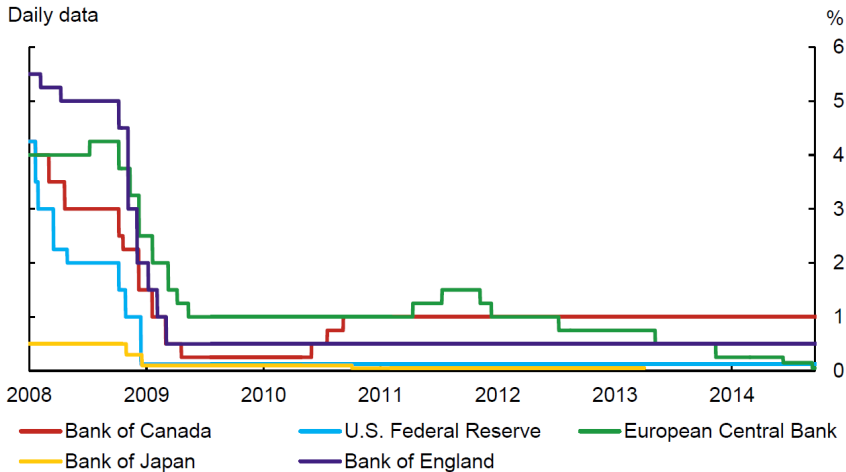
# Great Recession 2007-2009 and ZLB crisis in Canada

- Canada was not at the epicenter of Great Recession.
- Unlike the U.S. and Europe, the Canadian economy did not experience a subprime crisis in 2007-2008.
- However, the contagion spread through a number of transmission channels.
- For Canada, an important transmission channel was through a direct impact on foreign trade.

There is a popular saying “*When the U.S. sneezes, Canada catches a cold*”. But this time it went the other way around: it was the U.S. that caught the (subprime crisis) cold, and it was Canada that sneezed.

# Great Recession in Canada

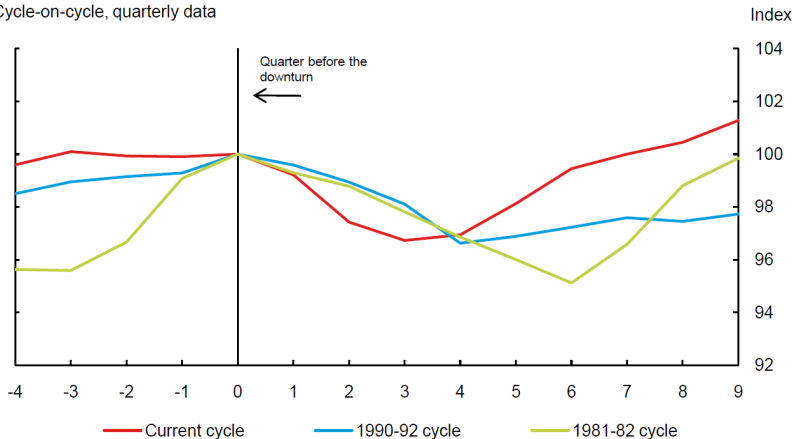
## A sharp decline in target interest rates



# Great Recession in Canada

## A faster drop and faster recovery in real GDP

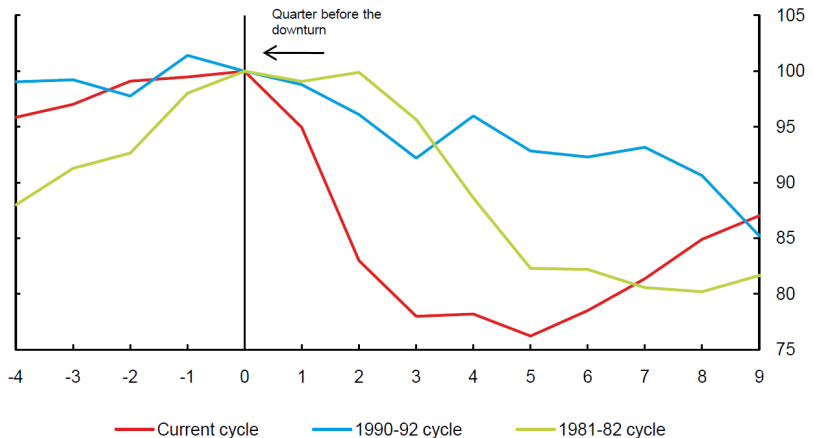
Cycle-on-cycle, quarterly data



# Great Recession in Canada

## A rapid drop in real fixed investment

Cycle-on-cycle, quarterly data

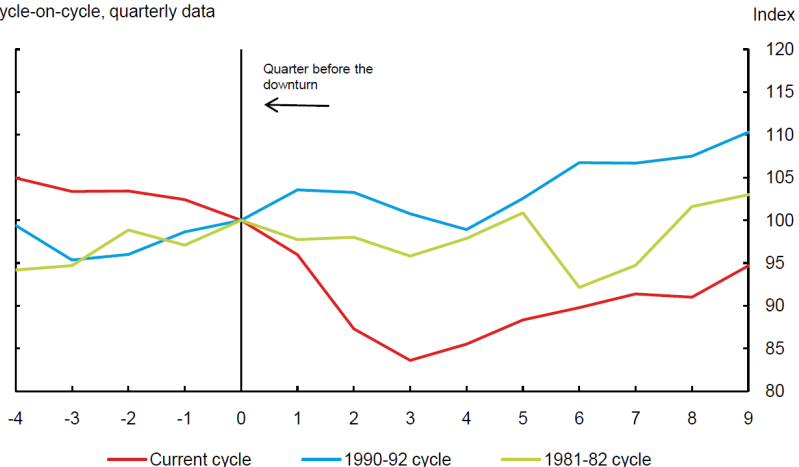


Sources: Statistics Canada and Bank of Canada calculations

# Great Recession in Canada

## A historic drop in real exports

Cycle-on-cycle, quarterly data



Sources: Statistics Canada and Bank of Canada calculations



# Full-scaled ToTEM model of Bank of Canada

- The Terms of Trade Economic Model (ToTEM) – the main projection and policy analysis model of the Bank of Canada.
- Small-open economy model.
- ToTEM includes 356 equations and unknowns  
⇒ It is too large for the existing global solution methods.

# A scaled-down version of ToTEM

- We construct a scaled-down version of ToTEM, which we call a “baby ToTEM” (bToTEM) model.
- bToTEM includes 49 equations and unknowns  
⇒ It is still a large-scale model.
- Two production sectors: final-good production and commodity production.
- Meaningful trade: final goods, commodities, imports.
- One representative household, with differentiated labour services.
- Taylor-type interest rate rule
- Six shocks, including exogenous rest-of-the-world (ROW) shocks.

# Differences between ToTEM and bToTEM

- *ToTEM*
  - 5 distinct production sectors (consumption goods and services, investment goods, government goods, noncommodity export goods, and commodities);
  - 4 sectors are identical except of parameters, while the commodity sector is different;
  - A separate economic model of the rest of the world (ROW);
  - 3 types of households (they differ in their saving opportunities);
  - 8 Phillips curves.
- *bToTEM*
  - The final-good production sector is identical in structure to the consumption goods and services sector of ToTEM;
  - Linear technologies for transforming the output of this sector into other types of output that correspond to the remaining ToTEM's sectors;
  - The ROW sector is modeled using exogenous processes for foreign variables;
  - All households are of the same type;
  - 3 Phillips curves.

# Production of final goods

## Two stages:

1. In the first stage, intermediate goods are produced by identical perfectly competitive firms from labour, capital, commodities, and imports.
2. In the second stage, a variety of final goods are produced by monopolistically competitive firms from the intermediate goods. The variety of final goods is then aggregated into the final consumption good.

# First stage of production

- Perfectly competitive firms produce an intermediate good:

$$Z_t^g = \left( \delta_l (A_t L_t)^{\frac{\sigma-1}{\sigma}} + \delta_k (u_t K_t)^{\frac{\sigma-1}{\sigma}} + \delta_{com} (COM_t^d)^{\frac{\sigma-1}{\sigma}} + \delta_m (M_t)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (1)$$

$L_t$ ,  $K_t$ ,  $COM_t^d$  = labour, capital and commodity inputs, resp.,  
 $M_t$  = imports,  $u_t$  = capital utilization,  $A_t$  = the level of labour-augmenting technology,

$$\log(A_t) = \varphi_a \log(A_{t-1}) + (1 - \varphi_a) \log(\bar{A}) + \xi_t^a. \quad (2)$$

- Capital depreciates according to the following law of motion

$$K_{t+1} = (1 - d_t) K_t + I_t, \quad (3)$$

where  $d_t$  is the depreciation rate, and  $I_t$  is investment.

- The depreciation rate increases with capital utilization:

$$d_t = d_0 + \bar{d} e^{\rho(u_t - 1)}. \quad (4)$$

## First stage of production (cont.)

- The firms incur a quadratic adjustment cost when adjusting the level of investment. The net output is given by

$$Z_t^n = Z_t^g - \frac{\chi_i}{2} \left( \frac{I_t}{I_{t-1}} - 1 \right)^2 I_t. \quad (5)$$

- The objective of the firms is to choose  $L_t$ ,  $K_{t+1}$ ,  $I_t$ ,  $COM_t$ ,  $M_t$ ,  $u_t$  in order to maximize profits

$$E_0 \sum_{t=0}^{\infty} \mathcal{R}_{0,t} (P_t^z Z_t^n - W_t L_t - P_t^{com} COM_t^d - P_t^i I_t - P_t^m M_t)$$

s.t. (1) – (5),

$\mathcal{R}_{t,t+j} = \beta^j (\lambda_{t+j}/\lambda_t) (P_t/P_{t+j}) =$  stochastic discount factor.

## Second stage of production

- A monopolistically competitive firm  $i \in [0, 1]$  produces a differentiated good

$$Z_{it} = \min \left( \frac{Z_{it}^n}{1 - s_m}, \frac{Z_{it}^{mi}}{s_m} \right),$$

$Z_{it}^n$  = intermediate good;  $Z_{it}^{mi}$  = manufactured input;  
 $s_m$  = a Leontief parameter.

- The differentiated goods  $Z_{it}$  are aggregated into the final good  $Z_t$  according to a CES function:

$$Z_t = \left( \int_0^1 Z_{it}^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

- Cost minimization implies

$$Z_{it} = \left( \frac{P_{it}}{P_t} \right)^{-\varepsilon} Z_t,$$

where  $P_t = \left( \int_0^1 P_{it}^{1-\varepsilon} di \right)^{\frac{1}{1-\varepsilon}}.$

## Second stage of production (cont.)

- The final good is used as the manufactured inputs by each of the monopolistically competitive firms.
- Two types of monopolistically competitive firms: rule-of-thumb firms of measure  $\omega$  and forward-looking firms of measure  $1 - \omega$ .
- Within each type, with probability  $\theta$  the firms index their price to the (time-varying) inflation target  $\bar{\pi}_t$ .
- The rule-of-thumb firms, which do not index their price in the current period, partially index their price:

$$P_{it} = (\pi_{t-1})^\gamma (\bar{\pi}_t)^{1-\gamma} P_{i,t-1}.$$



## Second stage of production (cont.)

- The optimizing forward-looking firms solve:

$$\max_{P_t^*} E_t \left\{ \sum_{j=0}^{\infty} \theta^j \mathcal{R}_{t,t+j} \left( \prod_{k=1}^j \bar{\pi}_{t+k} P_t^* Z_{i,t+j} \right. \right. \\ \left. \left. - (1 - s_m) P_{t+j}^z Z_{i,t+j} - s_m P_{t+j} Z_{i,t+j} \right) \right\}$$

subject to demand constraints

$$\text{s.t. } Z_{i,t+j} = \left( \frac{\prod_{k=1}^j \bar{\pi}_{t+k} P_t^*}{P_{t+j}} \right)^{-\varepsilon} Z_{t+j}.$$

# Relation between the first and second stages of production

- Production in the first and second stages are related as

$$Z_t^n = \int_0^1 Z_{it}^n di = (1 - s_m) \int_0^1 Z_{it} di = (1 - s_m) \Delta_t Z_t,$$

$$\Delta_t = \int_0^1 \left( \frac{P_{it}}{P_t} \right)^{-\varepsilon} di = \text{price dispersion}.$$

- Law of motion of the price dispersion:

$$\Delta_t = \theta \left( \frac{\bar{\pi}_t}{\pi_t} \right)^{-\varepsilon} \Delta_{t-1} + (1 - \theta) \omega \frac{(\pi_{t-1})^\gamma (\bar{\pi}_t)^{1-\gamma}}{\pi_t} \Delta_{t-1} + (1 - \theta) (1 - \omega) \left( \frac{P_t^*}{P_t} \right)^{-\varepsilon}.$$

- Investment goods and noncommodity exports are produced from the final goods according to linear technology,  $P_t^i = \iota_i P_t$  and  $P_t^{nc} = \iota_x P_t$ .

# Imports

- Intermediate imported goods  $M_{it}$  are bounded into the final imported good  $M_t$  according to

$$M_t = \left( \int_0^1 M_{it}^{\frac{\varepsilon_m - 1}{\varepsilon_m}} di \right)^{\frac{\varepsilon_m}{\varepsilon_m - 1}}.$$

- The demand for an intermediate imported good  $i$ :

$$M_{it} = \left( \frac{P_{it}^m}{P_t^m} \right)^{-\varepsilon_m} M_t,$$

where  $P_t^m = \left( \int_0^1 (P_{it}^m)^{1-\varepsilon_m} di \right)^{\frac{1}{1-\varepsilon_m}}.$

## Import (cont.)

- The optimizing forward-looking firms solves

$$\max_{P_t^{m*}} E_t \left[ \sum_{j=0}^{\infty} (\theta_m)^j \mathcal{R}_{t,t+j} \prod_{k=1}^j \bar{\pi}_{t+k} P_t^{m*} M_{i,t+j} - e_{t+j} P_{t+j}^{mf} M_{i,t+j} \right]$$

subject to demand constraints

$$M_{i,t+j} = \left( \frac{\prod_{k=1}^j \bar{\pi}_{t+k} P_t^{m*}}{P_{t+j}^m} \right)^{-\varepsilon_m} M_{t+j},$$

$P_t^{mf}$  = foreign price of imports;

$e_t$  = nominal exchange rate (domestic price of a unit of foreign currency).

# Foreign demand and foreign economy

- The foreign demand for Canadian noncommodity exports  $X_t^{nc}$  is given by the demand function

$$X_t^{nc} = \gamma^f \left( \frac{P_t^{nc}}{e_t P_t^f} \right)^{-\varphi} Z_t^f,$$

$P_t^{nc}/e_t$  = foreign price of noncommodity exports;

$P_t^f$  = foreign general price level;

$Z_t^f$  = foreign output.

- The balance of payments

$$\frac{e_t B_t^f}{R_t^f (1 + \kappa_t^f)} - e_t B_{t-1}^f = P_t^{nc} X_t^{nc} + P_t^{com} X_t^{com} - P_t^m M_t,$$

$B_t^f$  = domestic holdings of foreign-currency denominated bonds.;

$\kappa_t^f$  = risk premium.

## Foreign demand and foreign economy (cont.)

- The rest of the world is specified by three exogenous processes that describe the evolution of foreign variables.
- The foreign output  $Z_t^f$

$$\log \left( Z_t^f \right) = \varphi_{Zf} \log \left( Z_{t-1}^f \right) + (1 - \varphi_{zf}) \log \left( \bar{Z}^f \right) + \xi_t^{zf}.$$

- A foreign interest rate shock  $r_t^f$

$$\log \left( r_t^f \right) = \varphi_{rf} \log \left( r_{t-1}^f \right) + (1 - \varphi_{rf}) \log \left( \bar{r} \right) + \xi_t^{rf}.$$

- A foreign commodity price  $p_t^{comf}$  is

$$\log \left( p_t^{comf} \right) = \varphi_{comf} \log \left( p_{t-1}^{comf} \right) + (1 - \varphi_{comf}) \log \left( \bar{p}^{comf} \right) + \xi_t^{comf},$$

$\xi_t^{zf}, \xi_t^{rf}, \xi_t^{comf}$  = normally distributed random variables;

$\varphi_{Zf}, \varphi_{rf}, \varphi_{comf}$  = autocorrelation coefficients.

# Commodity production

- The commodities are produced from the final goods by representative, perfectly competitive domestic firms:

$$COM_t = (Z_t^{com})^{s_z} (A_t F)^{1-s_z} - \frac{\chi_{com}}{2} \left( \frac{Z_t^{com}}{Z_{t-1}^{com}} - 1 \right)^2 Z_t^{com},$$

$F$  = a fixed production factor (land).

Here, the second term is quadratic adjustment costs.

- The commodities are sold at the rest-of-the-world price adjusted by the nominal exchange rate

$$P_t^{com} = e_t P_t^{comf}.$$

- The commodities are sold domestically ( $COM_t^d$ ) or exported to the rest of the world ( $X_t^{com}$ )

$$COM_t = COM_t^d + X_t^{com}.$$

# Households

- The representative household's period utility function:

$$U_t = \frac{\mu}{\mu - 1} (C_t - \xi \bar{C}_{t-1})^{\frac{\mu-1}{\mu}} \exp \left( \frac{\eta(1-\mu)}{\mu(1+\eta)} \int_0^1 (L_{ht})^{\frac{\eta+1}{\eta}} dh \right) \eta_t^c,$$

$C_t$  = consumption of finished goods;

$\bar{C}_t$  = aggregate consumption;

$L_{ht}$  = labour service of type  $h$ ;

$\eta_t^c$  = a consumption demand shock.

- We assume

$$\log(\eta_t^c) = \varphi_c \log(\eta_{t-1}^c) + \xi_t^c,$$

$\xi_t^c$  = a normally distributed variable;  $\varphi_c$  = an autocorrelation coefficient.



## Households (cont.)

- The representative household solves

$$E_t \left[ \sum_{j=0}^{\infty} \beta^j U_{t+j} \right]$$

$$\text{s.t. } P_t C_t + \frac{B_t}{R_t} + \frac{e_t B_t^f}{R_t^f (1 + \kappa_t^f)} = B_{t-1} + e_t B_{t-1}^f + \int_0^1 W_{ht} L_{ht} dh + \Pi_t,$$

$B_t$  = holdings of domestic bonds;

$B_t^f$  = holdings of foreign-currency denominated bonds;

$\Pi_t$  are profits paid by the firms.

- To induce the stationarity of the model, we assume that the risk premium  $\kappa_t^f$  is

$$\kappa_t^f = \varsigma \left( \bar{b}^f - b_t^f \right),$$

$b_t^f = e_t B_t^f / \left( \pi_{t+1}^f P_t \bar{Y} \right)$  = normalized bond holdings; see Schmitt-Grohé and Uribe (2003).

# Wage setting

- Households;
- Aggregating firms (labor packers) – in perfect competition;
- Labour unions – in monopolistic competition.

# Labour packers

- The representative household supplies a variety of differentiated labour service  $L_{ht}$ ,  $h \in [0, 1]$  to the labour market.
- The differentiated labour service is aggregated by labor packers according to

$$L_t = \left( \int_0^1 L_{ht}^{\frac{\varepsilon_w - 1}{\varepsilon_w}} dh \right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}}.$$

- $L_t$  is used in the first stage of production.
- Cost minimization of the labor packer implies the demand

$$L_{ht} = \left( \frac{W_{ht}}{W_t} \right)^{-\varepsilon_w} L_t,$$

$$W_{ht} = \text{wage for labour of type } h; W_t \equiv \left( \int_0^1 W_{ht}^{1-\varepsilon_w} dh \right)^{\frac{1}{1-\varepsilon_w}}.$$

## Labour unions (cont.)

- Two types: rule-of-thumb (measure  $\omega_w$ ) and forward-looking (measure  $1 - \omega_w$ ) unions.
- Within each type, with probability  $\theta_w$  a union indexes wage to the inflation target  $\bar{\pi}_t$ ,  $W_{it} = \bar{\pi}_t W_{i,t-1}$ .
- The rule-of-thumb unions which do not index their wage set

$$W_{it} = (\pi_{t-1}^w)^{\gamma_w} (\bar{\pi}_t)^{1-\gamma_w} W_{i,t-1}.$$

- The forward-looking unions that do not index their wage solve:

$$\begin{aligned} \max_{W_t^*} E_t \left[ \sum_{j=0}^{\infty} (\beta \theta_w)^j U_{t+j} \right] \\ \text{s.t. } L_{h,t+j} = \left( \frac{\prod_{k=1}^j \bar{\pi}_{t+k} W_t^*}{W_{t+j}} \right)^{-\varepsilon_w} L_{t+j}, \\ P_{t+j} C_{t+j} = \prod_{k=1}^j \bar{\pi}_{t+k} W_t^* L_{h,t+j} dh + \dots \end{aligned}$$

# Monetary policy

- Taylor rule:

$$\begin{aligned} R_t &= \rho_r R_{t-1} + (1 - \rho_r) [\bar{R} + \rho_\pi (\pi_t - \bar{\pi}_t) + \rho_Y (\log Y_t - \log \bar{Y}_t)] + \eta_t^r \\ &\equiv \Phi_t, \end{aligned}$$

$\bar{Y}_t$  = potential output;  $\eta_t^r$  = interest rate shock,

$$\eta_t^r = \varphi_r \eta_{t-1}^r + \xi_t^r,$$

$\xi_t^r$  = a normally distributed variable;  $\varphi_r$  = autocorrelation coefficient.

- Potential output changes with productivity as

$$\log \bar{Y}_t = \varphi_{zf} \log \bar{Y}_{t-1} + (1 - \varphi_{zf}) \log \left( \frac{A_t \bar{Y}}{\bar{A}} \right).$$

- If an effective lower bound (ELB) is imposed on the nominal interest rate,  $R_t^{ELB}$ , then

$$R_t = \max \{ \Phi_t, R_t^{ELB} \}.$$

# Market clearing conditions

- Resource feasibility condition

$$Z_t = C_t + \iota_i I_t + \iota_x X_t^{nc} + Z_t^{com} + v_z Z_t.$$

- GDP

$$Y_t = C_t + I_t + X_t^{nc} + X_t^{com} - M_t + v_y Y_t.$$

- GDP deflator

$$P_t^y Y_t = P_t C_t + P_t^i I_t + P_t^{nc} X_t^{nc} + P_t^{com} X_t^{com} - P_t^m M_t + v_y p_t^y Y_t.$$

# A list of model's variables

| #  | variable                              | notation    |
|----|---------------------------------------|-------------|
| 1  | productivity                          | $A_t$       |
| 2  | labour input                          | $L_t$       |
| 3  | capital input                         | $K_t$       |
| 4  | investment                            | $I_t$       |
| 5  | commodities used domestically         | $COM_t^d$   |
| 6  | import                                | $M_t$       |
| 7  | capital utilization                   | $u_t$       |
| 8  | capital depreciation                  | $d_t$       |
| 9  | gross production of intermediate good | $Z_t^g$     |
| 10 | net production of intermediate good   | $Z_t^n$     |
| 11 | total production                      | $Z_t$       |
| 12 | consumption                           | $C_t$       |
| 13 | GDP                                   | $Y_t$       |
| 14 | marginal utility of consumption       | $\lambda_t$ |
| 15 | inflation                             | $\pi_t$     |
| 16 | real marginal cost                    | $rmc_t$     |
| 17 | consumption Phillips curve term       | $F_{1t}$    |
| 18 | consumption Phillips curve term       | $F_{2t}$    |
| 19 | price dispersion                      | $\Delta_t$  |

## A list of model's variables (cont.)

| #  | variable                              | notation      |
|----|---------------------------------------|---------------|
| 20 | imported good inflation               | $\pi_t^m$     |
| 21 | imports Phillips curve term           | $F_{1t}^m$    |
| 22 | imports Phillips curve term           | $F_{2t}^m$    |
| 23 | inflation target                      | $\bar{\pi}_t$ |
| 24 | real price of import                  | $p_t^m$       |
| 25 | real exchange rate                    | $s_t$         |
| 26 | nominal interest rate                 | $R_t$         |
| 27 | interest rate shock process           | $\eta_t^r$    |
| 28 | real price of intermediate good       | $p_t^z$       |
| 29 | real wage                             | $w_t$         |
| 30 | real price of commodities             | $p_t^{com}$   |
| 31 | marginal product of capital           | $MPK_t$       |
| 32 | interest rate on capital              | $R_t^k$       |
| 33 | Tobin's Q                             | $q_t$         |
| 34 | real price of investment              | $p_t^i$       |
| 35 | foreign-currency price of commodities | $p_t^{comf}$  |
| 36 | foreign real interest rate            | $r_t^f$       |



## A list of model's variables (cont.)

| #  | variable                                 | notation     |
|----|------------------------------------------|--------------|
| 37 | interest premium on foreign bonds        | $\kappa_t^f$ |
| 38 | holdings of foreign bonds in real terms  | $b_t^f$      |
| 39 | non-commodity export                     | $X_t^{nc}$   |
| 40 | export of commodities                    | $X_t^{com}$  |
| 41 | foreign demand                           | $Z_t^f$      |
| 42 | total commodities produced               | $COM_t$      |
| 43 | final goods used in commodity production | $Z_t^{com}$  |
| 44 | wage inflation                           | $\pi_t^w$    |
| 45 | wage Phillips curve term                 | $F_{1t}^w$   |
| 46 | wage Phillips curve term                 | $F_{2t}^w$   |
| 47 | optimal wage                             | $w_t^*$      |
| 48 | wage dispersion                          | $\Delta_t^w$ |
| 49 | foreign price of import                  | $p_t^{mf}$   |
| 50 | GDP deflator                             | $p_t^y$      |
| 51 | price of non-commodity export            | $p_t^{xz}$   |
| 52 | consumption demand shock process         | $\eta_t^c$   |
| 53 | potential GDP                            | $\bar{Y}_t$  |

# Calibration

- 61 parameters to calibrate.
- Whenever possible, we use the same parameters as in ToTEM.
- We choose the remaining parameters to reproduce observations on the Canadian economy.
- We target the ratios of the following variables to nominal GDP:
  - consumption,
  - investment,
  - noncommodity export,
  - commodity export,
  - import,
  - total commodities,
  - labor input.

# Parameters in endogenous model's equations

| Parameter                          | Symbol         | Value  | Source        |
|------------------------------------|----------------|--------|---------------|
| Rates                              |                |        |               |
| – real interest rate               | $\bar{r}$      | 1.0076 | ToTEM         |
| – discount factor                  | $\beta$        | 0.9925 | ToTEM         |
| – inflation target                 | $\bar{\pi}$    | 1.005  | ToTEM         |
| – nominal interest rate            | $\bar{R}$      | 1.0126 | ToTEM         |
| – ELB on the nominal interest rate | $R^{ELB}$      | 1.01   | fixed         |
| Output production                  |                |        |               |
| – CES elasticity of substitution   | $\sigma$       | 0.5    | ToTEM         |
| – CES labor share parameter        | $\delta_L$     | 0.249  | calibrated    |
| – CES capital share parameter      | $\delta_K$     | 0.575  | calibrated    |
| – CES commodity share parameter    | $\delta_{COM}$ | 0.0015 | calibrated    |
| – CES import share parameter       | $\delta_M$     | 0.0287 | calibrated    |
| – investment adjustment cost       | $\chi_I$       | 20     | calibrated    |
| – fixed depreciation rate          | $d_0$          | 0.0054 | ToTEM         |
| – variable depreciation rate       | $\bar{d}$      | 0.0261 | ToTEM         |
| – depreciation semielasticity      | $\rho$         | 4.0931 | calibrated    |
| – real investment price            | $\iota_I$      | 1.2698 | ToTEM         |
| – real noncommodity export price   | $\iota_X$      | 1.143  | ToTEM         |
| – labour productivity              | $\bar{A}$      | 100    | normalization |

# Parameters in endogenous model's equations

| Parameter                                         | Symbol          | Value  | Source |
|---------------------------------------------------|-----------------|--------|--------|
| Price setting parameters for consumption          |                 |        |        |
| – probability of indexation                       | $\theta$        | 0.75   | ToTEM  |
| – RT indexation to past inflation                 | $\gamma$        | 0.0576 | ToTEM  |
| – RT share                                        | $\omega$        | 0.4819 | ToTEM  |
| – elasticity of substitution of consumption goods | $\varepsilon$   | 11     | ToTEM  |
| – Leontieff technology parameter                  | $s_m$           | 0.6    | ToTEM  |
| Price setting parameters for imports              |                 |        |        |
| – probability of indexation                       | $\theta^m$      | 0.8635 | ToTEM  |
| – RT indexation to past inflation                 | $\gamma^m$      | 0.7358 | ToTEM  |
| – RT share                                        | $\omega^m$      | 0.3    | ToTEM  |
| – elasticity of substitution of imports           | $\varepsilon^m$ | 4.4    | ToTEM  |
| Price setting parameters for wages                |                 |        |        |
| – probability of indexation                       | $\theta^w$      | 0.5901 | ToTEM  |
| – RT indexation to past inflation                 | $\gamma^w$      | 0.1087 | ToTEM  |
| – RT share                                        | $\omega^w$      | 0.6896 | ToTEM  |
| – elasticity of substitution of labour service    | $\varepsilon^w$ | 1.5    | ToTEM  |

# Parameters in endogenous model's equations

| Parameter                                 | Symbol           | Value  | Source     |
|-------------------------------------------|------------------|--------|------------|
| Household utility                         |                  |        |            |
| – consumption habit                       | $\xi$            | 0.9396 | ToTEM      |
| – consumption elasticity of substitution  | $\mu$            | 0.8775 | ToTEM      |
| – wage elasticity of labor supply         | $\eta$           | 0.0704 | ToTEM      |
| Taylor rule                               |                  |        |            |
| – interest rate persistence parameter     | $\rho_r$         | 0.83   | ToTEM      |
| – interest rate response to inflation gap | $\rho_\pi$       | 4.12   | ToTEM      |
| – interest rate response to output gap    | $\rho_y$         | 0.4    | ToTEM      |
| Other                                     |                  |        |            |
| – capital premium                         | $\kappa^k$       | 0.0674 | calibrated |
| – exchange rate persistence parameter     | $\varkappa$      | 0.1585 | ToTEM      |
| – foreign commodity price                 | $\bar{p}^{COMf}$ | 1.6591 | ToTEM      |
| – foreign import price                    | $\bar{p}^{Mf}$   | 1.294  | ToTEM      |
| – risk premium response to debt           | $\varsigma$      | 0.0083 | calibrated |
| – foreign demand elasticity               | $\phi$           | 0.4    | calibrated |
| – elasticity in commodity production      | $s_z$            | 0.8    | calibrated |
| – land                                    | $F$              | 0.1559 | calibrated |
| – share of other components of output     | $v_Z$            | 0.7651 | calibrated |
| – share of other components of GDP        | $v_Y$            | 0.311  | calibrated |

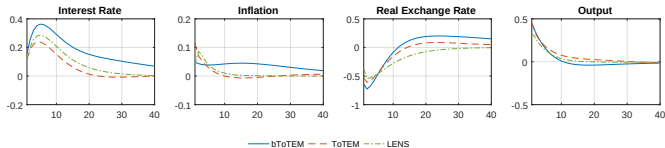
# Parameters in exogenous model's equations

| Parameter                                      | Symbol            | Value  | Source     |
|------------------------------------------------|-------------------|--------|------------|
| Shock persistence                              |                   |        |            |
| – persistence of interest rate shock           | $\varphi_R$       | 0.25   | ToTEM      |
| – persistence of productivity shock            | $\varphi_A$       | 0.9    | fixed      |
| – persistence of consumption demand shock      | $\varphi_C$       | 0      | fixed      |
| – persistence of foreign output shock          | $\varphi_{Zf}$    | 0.9    | fixed      |
| – persistence of foreign commodity price shock | $\varphi_{pCOMf}$ | 0.87   | calibrated |
| – persistence of foreign interest rate shock   | $\varphi_{rf}$    | 0.88   | calibrated |
| Shock volatilities                             |                   |        |            |
| – volatility of interest rate shock            | $\sigma_R$        | 0.0006 | calibrated |
| – volatility of productivity shock             | $\sigma_A$        | 0.0067 | calibrated |
| – volatility of consumption demand shock       | $\sigma_C$        | 0.0001 | fixed      |
| – volatility of foreign output shock           | $\sigma_{Zf}$     | 0.0085 | calibrated |
| – volatility of foreign commodity price shock  | $\sigma_{pCOMf}$  | 0.0796 | calibrated |
| – volatility of foreign interest rate shock    | $\sigma_{rf}$     | 0.002  | calibrated |

# A comparison of bToTEM to ToTEM and LENS

- The Bank of Canada uses a first-order perturbation method to solve ToTEM.
- For ToTEM, we use IRIS – open-source software used by the Bank of Canada for macroeconomic modeling.
- For bToTEM, we use IRIS Toolbox, as well as Dynare.
- We checked that the IRIS and Dynare packages produce indistinguishable numerical solutions for bToTEM.
- Also, we include for comparison the LENS model – another model of the Bank of Canada.
- LENS is a semistructural model.
- Both, the ToTEM and LENS models, include more shocks than bToTEM:
  - 52 shocks in ToTEM and 98 shocks in LENS.

# Impulse response to a consumption demand shock





# Understanding the difference between the bToTEM and ToTEM

- Consider a linearized version of the Phillips curve in the two models

$$\hat{\pi}_t = (1 - \theta) \gamma \omega \tilde{\phi}^{-1} \hat{\pi}_{t-1} + \beta \theta \tilde{\phi}^{-1} E[\hat{\pi}_{t+1}] + \tilde{\lambda} r\hat{m}c_t + \varepsilon_t^p,$$

$r\hat{m}c_t$  = real marginal cost;

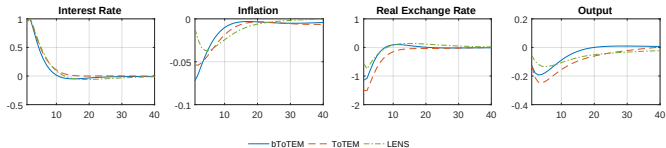
$\varepsilon_t^p$  = weighted average of the inflation target and the deviation of markup from the steady state;

$\theta, \gamma, \omega$  are the price stickiness parameters;

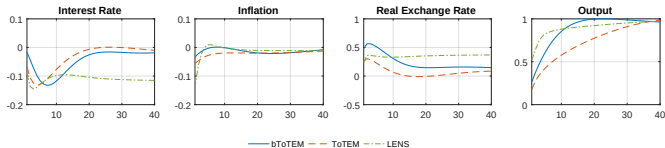
$\tilde{\phi}$  and  $\tilde{\lambda}$  = parameters that are identical in ToTEM and bToTEM (defined in Dorich et al., 2013).

- We conclude that inflation dynamics are driven by differing  $r\hat{m}c_t$ .
- In ToTEM, a consumption shock triggers a reallocation of inputs into the consumption good sector from the other four sectors. With adjustment costs, this raises  $r\hat{m}c_t$ .
- In contrast, in bToTEM, there is one production sector and there are no input adjustment costs. The responses and decays of the  $r\hat{m}c_t$  are far less pronounced.

# Impulse response to an interest rate shock



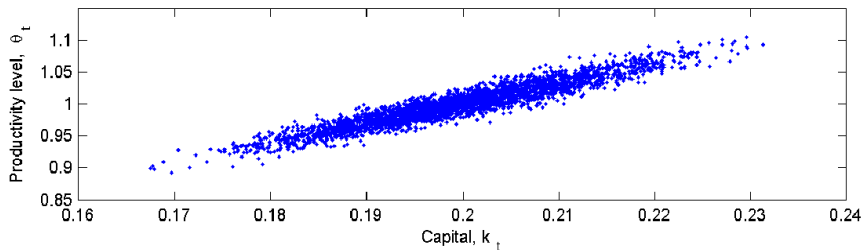
# Impulse response to a permanent productivity shock



# bToTEM: a serious challenge for global methods

- The models like bToTEM has not been yet studied in the literature using global methods.
- bToTEM contains 21 state variables (6 exogenous and 15 endogenous ones)  $\implies$  *curse of dimensionality*.
- Moreover, the bToTEM's equations are complex and require the use of numerical solvers.
- To solve the bToTEM model, we ameliorate the curse of dimensionality by focusing on the ergodic set of the model.

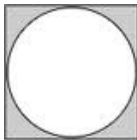
# Advantages of ergodic set methods



- Ergodic set is constructed by stochastic simulation.
- "Grid" is adaptive: we solve the model only in the high-probability area of the state space—this is where the solution "lives"!

## Reduction in cost in a 2-dimensional case

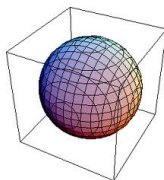
- How much can we save on cost using the ergodic-set domain comparatively to the hypercube domain?
- Suppose the ergodic set is a circle (it was an ellipse in the figure).
- In the 2-dimensional case, a circle inscribed within a square occupies about 79% of the area of the square.
- The reduction in cost is proportional to the shaded area in the figure.



- It does not seem to be a large gain.

# Reduction in cost in a d-dimensional case

- In a 3-dimensional case, the gain is larger (a volume of a sphere of diameter 1 is 52% of the volume of a cube of width 1)



- In a  $d$ -dimensional case, the ratio of a hypersphere's volume to a hypercube's volume

$$\mathcal{V}^d = \begin{cases} \frac{(\pi/2)^{\frac{d-1}{2}}}{1 \cdot 3 \cdot \dots \cdot d} & \text{for } d = 1, 3, 5, \dots \\ \frac{(\pi/2)^{\frac{d}{2}}}{2 \cdot 4 \cdot \dots \cdot d} & \text{for } d = 2, 4, 6, \dots \end{cases}$$

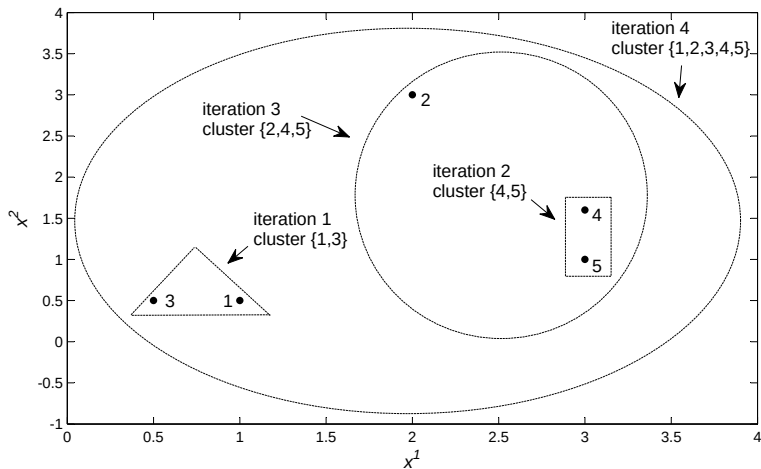
- $\mathcal{V}^d$  declines very rapidly with dimensionality of state space. When  $d = 10 \Rightarrow \mathcal{V}^d = 3 \cdot 10^{-3}$  (0.3%). When  $d = 30 \Rightarrow \mathcal{V}^d = 2 \cdot 10^{-14}$ .
- We face a tiny fraction of cost we would have faced on the hypercube.

# Cluster grid algorithm (CGA)

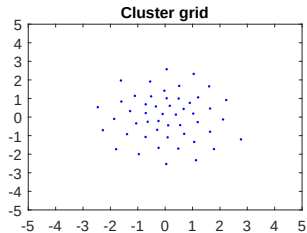
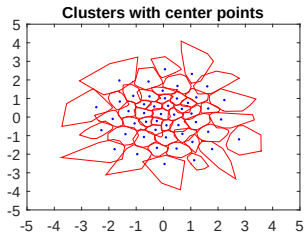
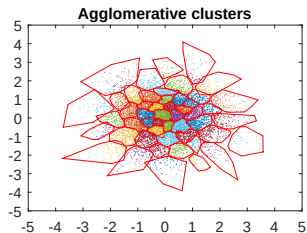
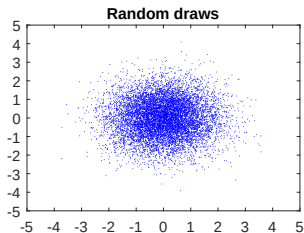
- CGA is an ergodic set algorithm of Maliar and Maliar (QE, 2015).
- It is a projection-style global solution method that uses adaptive grid:
  - the model is solved only in the area of the state space visited in simulation
- The essence of CGA is clustering methods from machine learning analysis.
  - Crude simulation produces points that are located close to one another; many points are redundant.
  - Clustering methods reduce the number of redundant points.
- CGA can accurately solve models with dozens of state variables.



# Example: agglomerative hierarchical clustering algorithm



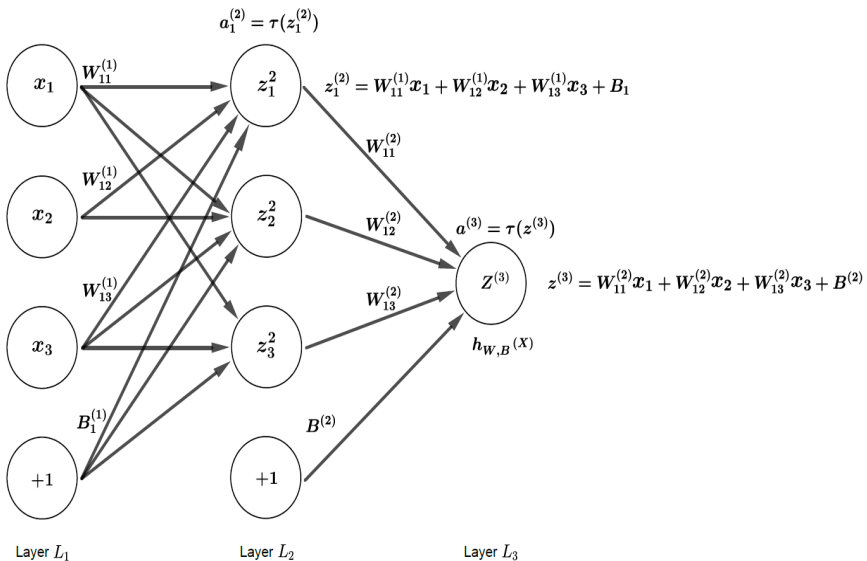
# Example: Construction of a cluster grid



# Deep learning

- In the 2017 version of the paper, we constructed nonlinear solutions using second-order polynomial approximation.
- But second-degree polynomial is not flexible enough to accurately approximate highly non-linear models like bToTEM.
- The difference between perturbation and global solutions cannot be too large as both of them are built using the same quadratic approximation function.
- To approximate nonlinearities more accurately, one needs more flexible approximating functions.
- In the new 2019 version, we introduce neural networks and deep learning techniques to produce more flexible approximations.

# Artificial neural networks



## Artificial neural networks

- $x = (x_0, x_1, \dots, x_n)'$  and
- $z_1^{(2)} = W_{11}^{(1)} x_1 + \dots + W_{1n}^{(1)} x_n + B^{(1)}$ .
- $W_{jk}^{(l)}$  = weights that control mapping from layer  $l$  to layer  $l + 1$ .
- $\tau$  = activation function (computation by a neuron).
- $a^{(l)}$  = activated output in layer  $l$ .
- $h_{W,B} (x^{(i)})$  = hypothesis about our relationship between  $x$ 's and  $y$ 's.
- Cost function  $J(W, B)$  allows us to evaluate how different  $h_{W,B} (x^{(i)})$  is from  $y$ .
- To prevent overfitting,  $J(W, B)$  may include penalty on large values of  $W_{j,k}^{(l)}$ .
- $\min_{W,B} J(W, B)$ . Need to compute  $\frac{\partial}{\partial W_{ij}^{(l)}} J(W, B)$ . Use backpropagation to compute gradient of each layer.
- Use a mean-squared error (MSE):

$$J(W, B) = \frac{1}{m} \sum_{i=1}^m \left( h_{W,B} (x^{(i)}) - y_k^{(i)} \right)^2 + \dots$$

# Understanding the role of nonlinearities

- i). **(Uncertainty effect)**. *Linearized solutions do not depend on the degree of volatility  $\sigma$ , as do nonlinear solutions.*
  - ii). **(High-order effect)**. *Linearization method neglects high-order polynomial terms, unlike more flexible nonlinear solutions.*
  - iii). **(Solution-domain effect)**. *Perturbation (local) solutions are constructed to be accurate in a deterministic steady state, and their accuracy can deteriorate dramatically when deviating from the steady state, in particular, in the area of ELB.*
- Let's look at these three effects for perturbation solutions,

$$g(x, \sigma) \approx \underbrace{g(\bar{x}, 0) + g_x(\bar{x}, 0)(x - \bar{x})}_{\text{1st-order perturbation solution}} + \underbrace{\frac{g_{\sigma\sigma}(\bar{x}, 0)\sigma^2}{2}}_{\text{Uncertainty effect}} + \underbrace{\frac{g_{xx}(\bar{x}, 0)(x - \bar{x})^2}{2}}_{\text{High-order effect}}$$

- The second-order perturbation method addresses i) and ii) by the terms  $\frac{1}{2}g_{\sigma\sigma}(\bar{x}, 0)\sigma^2$  and  $\frac{1}{2}g_{xx}(\bar{x}, 0)(x - \bar{x})^2$ , respectively. Plain perturbation methods do not address iii) but IRIS and OccBin do.
- Our DL solution method addresses i) - iii) more conclusively than the perturbation methods by using more flexible neural network approximation and global solution domain.

# Experiment 1: Foreign-driven recession

- The U.S. is the main Canadian trade partner (around 75% of Canadian exports goes to the US).
- In 2008, the Canadian economy experienced a huge 16% drop in exports.
- In 2009–2010, the Bank of Canada targeted the overnight interest rate at 0.25% (the lower bound).

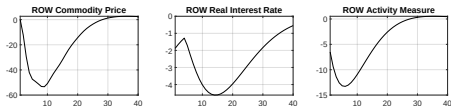
# Experiment 1: Exogenous ROW shocks

An important question is how to calibrate the rest of the world (ROW) sector in the bToTEM model since foreign financial crisis affects not just foreign demand but also foreign prices and foreign interest rates.

- we use ToTEM to produce impulse responses for 3 ROW variables (interest rate, commodity price, output).
- we use them as exogenous shocks in the bToTEM model.

The ROW shocks generated by ToTEM is in line with the data.

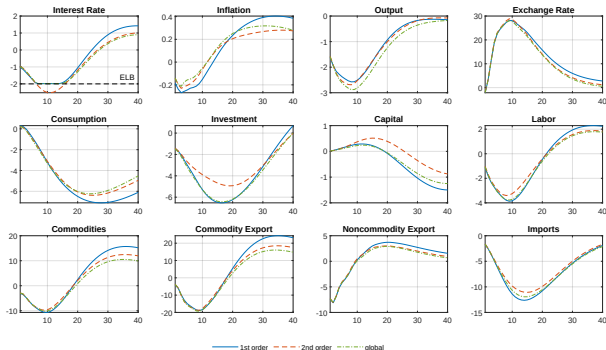
- The ROW commodity price shock matches a 50 percent decline in the global commodity price index documented by the IMF.
- The ROW real interest rate shock matches a 5 percent peak-to-trough decline in the effective U.S. federal funds rate.
- The ROW activity measure shock is about 18 percent decline in the foreign activity measure estimated by the Bank of Canada.





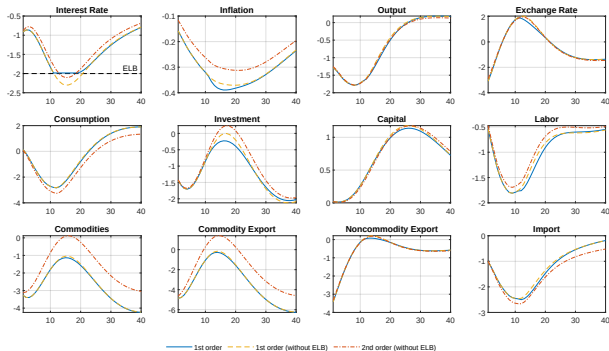
# Experiment 1: Foreign-driven recession

Linear local, quadratic local and global solutions



# Experiment 1: Domestic-driven recession

Linear local, quadratic local and global solutions



# Approximation errors

Residuals in the model's equations on the simulated path, log 10 units

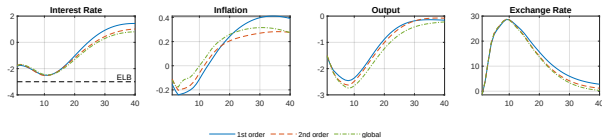
|             | Maximum error |           |           | Average error |           |           |
|-------------|---------------|-----------|-----------|---------------|-----------|-----------|
|             | Local         | Local     | Global    | Local         | Local     | Global    |
|             | 1st order     | 2nd order | 2nd order | 1st order     | 2nd order | 2nd order |
| $R_t$       | -3.83         | -3.84     | -5.07     | -4.13         | -4.53     | -5.74     |
| $\pi_t$     | -4.40         | -3.81     | -4.38     | -4.65         | -4.49     | -5.08     |
| $s_t$       | -2.38         | -1.96     | -2.51     | -3.16         | -2.62     | -3.24     |
| $Y_t$       | -2.59         | -3.27     | -3.42     | -2.78         | -3.90     | -4.16     |
| $C_t$       | -3.19         | -3.13     | -3.94     | -3.44         | -3.79     | -4.74     |
| $I_t$       | -3.01         | -3.39     | -3.51     | -3.79         | -4.01     | -4.09     |
| $X_t^{com}$ | -1.75         | -2.32     | -2.73     | -1.97         | -2.88     | -3.43     |
| $X_t^{nc}$  | -2.78         | -2.36     | -2.91     | -3.56         | -3.01     | -3.64     |
| $M_t$       | -2.17         | -2.90     | -3.17     | -2.41         | -3.53     | -3.88     |
| Average     | -2.76         | -2.81     | -3.41     | -3.20         | -3.45     | -4.11     |
| Max         | -1.43         | -1.44     | -2.09     | -1.92         | -2.08     | -2.75     |

## Experiment 2: Higher inflation target

- For the last 25 years, the Bank of Canada adhered to the inflation targeting, however, every three to five years it revises their inflation-control framework.
- Current inflation target is 2%.
- A higher inflation target could be beneficial by reducing frequency and severity of ELB episodes.
- Kryvtsov and Mendes (2015), Dorich et al. (2017).
- We use the bToTEM model to assess the effects of an increase in the inflation target from 2% to 3%.

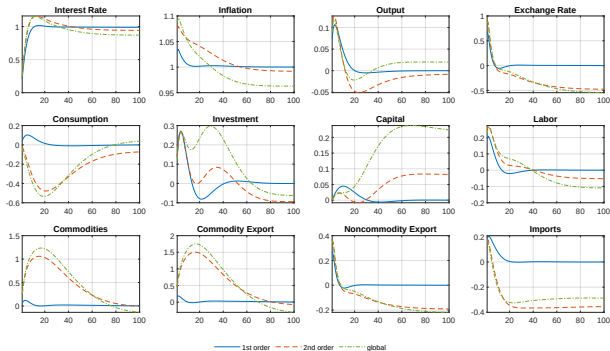
## Experiment 2: Higher inflation target

A 3% inflation target could prevent ELB episodes similar to the 2009-10 episode



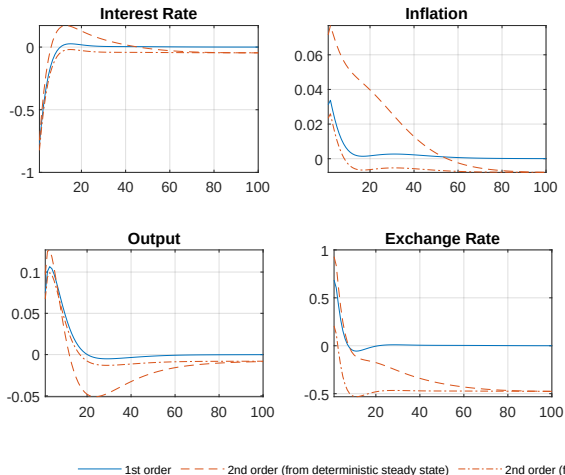
## Experiment 2: Higher inflation target

Transition from the deterministic steady state with the inflation target of 2%



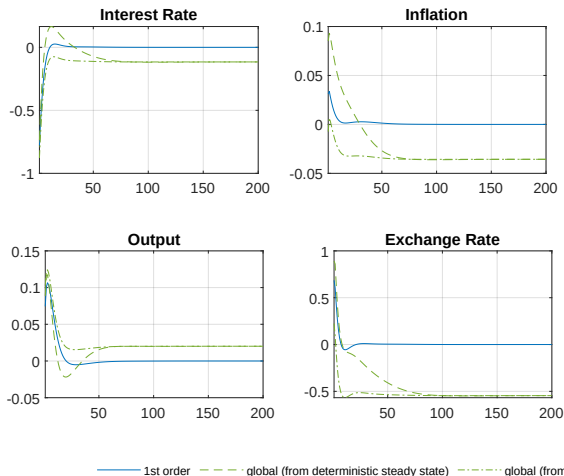
## Experiment 2: Higher inflation target

Transition from the different states with the target of 2%, 2nd order solution



## Experiment 2: Higher inflation target

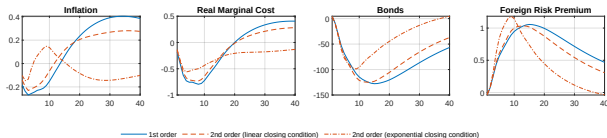
Transition from the different states with the target of 2%, global solution





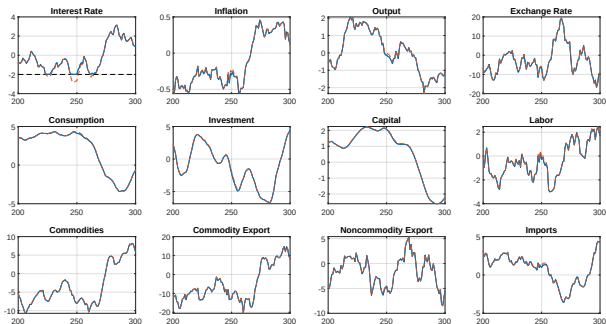
# Closing condition

- We compare the closing condition in an exponential form, used in Schmitt-Grohé and Uribe (2003)  $\kappa_t^f = \varsigma \left[ \exp \left( \bar{b}^f - b_t^f \right) - 1 \right]$  and the linear form  $\kappa_t^f = \varsigma \left( \bar{b}^f - b_t^f \right)$ .
- The importance of stationarity condition in the bToTEM model is surprising, given a well-known finding of Schmitt-Grohé and Uribe (2003) that the closing condition used does not significantly affect the implications of open-economy models.
- However, their analysis is obtained for linearized solutions in which the exponential and linear conditions coincide exactly but such conditions are different in our nonlinear analysis.



# ELB irrelevance

- In bToTEM, the probability of reaching ELB in bToTEM is large (about 8 percent) and the average duration of ELB episodes is 5 quarters but its effect is very small.
- In Debartoli et al. (2019), the ELB constraint is irrelevant because of the availability of unconventional monetary policies (forward guidance, quantitative easing, etc.) while bToTEM, it is irrelevant due to the presence of the rule-of-thumb agents introduced to deal with the forward guidance puzzle.



# Conclusion

- Large-scale central banking models like ToTEM and bToTEM are analyzed by linearization methods.
- Our novel DL algorithm combines supervised and unsupervised learning to construct an accurate global fully nonlinear solution.
- What is the value added of deep learning for telling the Canadian ELB tale?
- It is fair to say that we could have discovered the ELB contagion mechanism by using linearization-based methods.
- But we would not know how reliable our linear solution is, and we would miss some dramatic effects of nonlinearities on the solution.
- In particular, we would overlook the uncertainty effect.
- Furthermore, we would miss the nonlinearity effects associated with the closing condition.
- On the other hand, the nonlinearity associated with the ELB turned out to be of a lesser importance than we expected.

# Conclusion (cont.)

- The bToTEM model constructed in the paper is a useful alternative model to the Bank of Canada.
- While the full-scale ToTEM is not yet feasible for global nonlinear methods, bToTEM can be solved nonlinearly and its accuracy can be assessed.
- In addition to the Bank of Canada, our deep learning analysis can be useful to all users of large-scale models, including researchers, central banks and government agencies who can benefit from our methodology of calibrating, solving, and simulating large-scale macroeconomic models, as well as designing nontrivial policy experiments within such models.